Differentially Private Stochastic Gradient Descent (Abadi et al. 2016)

Clipping per-sample gradient norms and injecting calibrated Gaussian noise during stochastic gradient descent (DP-SGD) provides provable (epsilon, delta)-differential privacy for deep neural networks.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Bounding the influence of individual training examples via per-sample gradient clipping to maximum L_2 norm C and adding calibrated Gaussian noise N(0, σ^2 C^2 I) to mini-batch gradients guarantees (ε, \delta)-differential privacy for deep neural network training under the Moments Accountant framework.

EVIDENCE AND CAUSAL ASSESSMENT

Mathematical tracking of higher-order moments of privacy loss random variables (the Moments Accountant) establishes tight composition bounds over thousands of training steps, directly enabling non-convex deep learning with rigorous privacy bounds.

EFFECT SIZE AND PERSISTENCE

DP-SGD maintains neural network accuracy within modest degradations (e.g., 5–10% accuracy gap on complex vision/language tasks) while providing mathematical protection against membership inference and model inversion attacks for the entire operational life of the weights.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%). Gradient clipping disproportionately attenuates gradients from under-represented minority classes or rare examples, causing DP-SGD to exhibit disparate impact on model utility across class distributions.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Training with DP-SGD does not eliminate utility degradation; strict privacy guarantees (ε < 2) require substantial compute overhead and typically incur notable accuracy reductions relative to unconstrained SGD.

WHAT ACTION IS REASONABLY JUSTIFIED

Use DP-SGD and the Moments Accountant when training deep neural networks on sensitive user records, medical images, or confidential institutional communications.

VERIFIED SOURCES & DOI IDENTIFIERS

- Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., & Zhang, L. (2016). Deep Learning with Differential Privacy. Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, 308-318. https://doi.org/10.1145/2976749.2978318